

JavaScript Complete Guide

Variables and Data Types

JavaScript has var, let, and const for variable declaration. let is block-scoped and preferred over var. const creates immutable bindings. Data types include string, number, boolean, null, undefined, symbol, and object. Use typeof operator to check data types at runtime.

Functions and Closures

Functions are first-class citizens in JavaScript. Arrow functions provide concise syntax with lexical this binding. Closures allow functions to access variables from their outer scope even after the outer function returns. Higher-order functions take or return other functions.

Promises and Async/Await

Promises represent eventual completion or failure of async operations. Chain `.then()` for success and `.catch()` for errors. `async/await` makes async code look synchronous. `Promise.all` runs multiple promises in parallel. Always handle promise rejections to avoid unhandled errors.

ES6+ Features

ES6 introduced destructuring, spread operator, template literals, and default parameters. Classes provide syntactic sugar over prototype-based inheritance. Modules use import/export for code organization. Optional chaining (?.) safely accesses nested properties.

Error Handling

Use `try/catch/finally` for synchronous error handling. Throw custom `Error` objects with meaningful messages. In async code, use `try/catch` with `await` or `.catch()` with promises. Create custom error classes extending `Error` for specific error types.